R3.1 Proton transfer reactions

Reactions of acids

Learning outcomes

By the end of this section you should be able to formulate equations for the reactions between acids and metal oxides, metal hydroxides, hydrogencarbonates, carbonates, ammonia and amines.

Acids are often thought of in a negative way like in the case of acid rain, but there are many acids that are helpful to us. Can you think of any?

Two examples are stomach acid is needed to break down the food we eat to allow for chemical digestion to take place (**Figure 1**) and acidic cleaning products that are used to kill bacteria.

Some of these reactions are explored in this section.

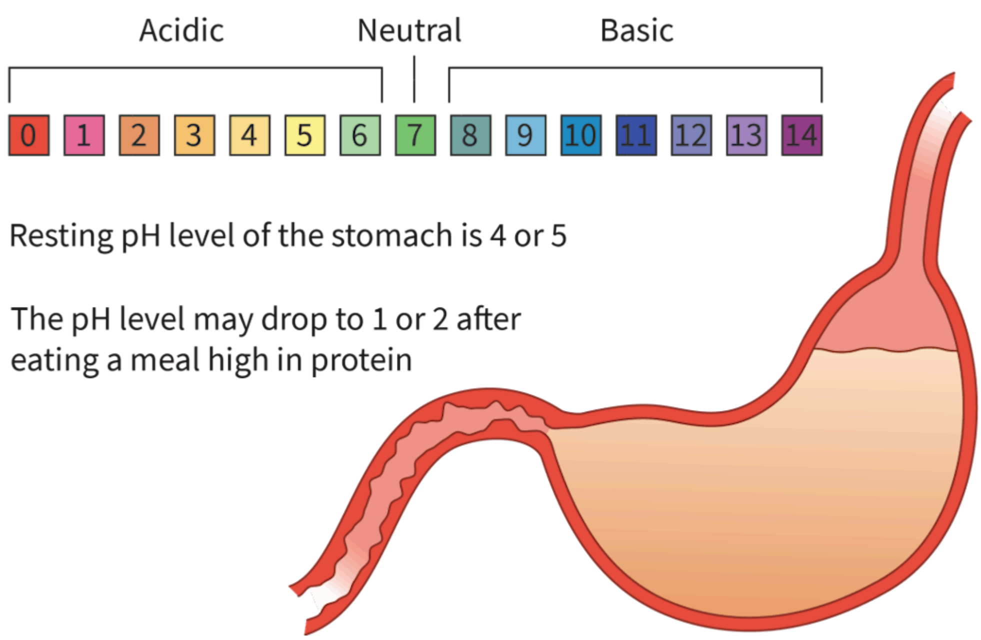

Figure 1. pH levels of the stomach before and after a meal.

More information for figure 1

The image is a diagram featuring an illustration of a stomach and a pH scale. The pH scale at the top is segmented into acidic, neutral, and basic regions, ranging from 0 to 14. The numbers 0 through 6, colored in gradients of red to yellow, represent acidic pH levels, with 0 being highly acidic. Number 7 is marked as neutral and is colored green. Numbers 8 through 14, colored in shades of blue to purple, represent basic pH levels.

Below the pH scale, two statements explain the pH levels concerning the stomach. The first statement indicates that the resting pH level of the stomach is 4 or 5, which falls into the acidic range of the scale. The second statement explains that the pH level may drop to 1 or 2 after consuming a high-protein meal, indicating increased acidity. The illustration of the stomach is shown beneath these statements, highlighting its shape and location in the digestive system.

[Generated by AI]

What happens to the pH of an acid (low pH) as it reacts with a base (high pH)?

The pH will move towards the middle, closer to neutral. Consequently, these reactions are known as neutralisation reactions.

In this section we will look into the reactions of acids with these bases:

- metal oxides and metal hydroxides
- hydrogencarbonates and carbonates
- ammonia and amines.

Study skills

Recall from section R3.1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/br%C3%B8nsted-lowry-acids-and-bases-id-43485/) that the distinction between the terms ‘base’ and ‘alkali’ should be understood:

Bases that are insoluble in water are simply called bases. Examples of bases include the metal oxides calcium oxide and copper oxide.

Bases that are soluble in water are known as alkalis. Examples of alkalis include the group 1 metal hydroxides, sodium hydroxide and lithium hydroxide.

Reaction with metal oxides

Acids undergo neutralisation reactions with metal oxides to produce a salt and water. This is a displacement reaction that will form a predictable set of products – the ionic salt and water. The reaction involves:

- The metal cation (M^+) in the oxide bonding with the anion (A^-) in the acid to form the salt (MA).
- The hydrogen (H^+) in the acid bonding with the oxygen (O) in the base to form water (H_2O).

The general word equation is given below, together with some specific examples:

$$\text{metal oxide} + \text{acid} \rightarrow \text{salt and water}$$
$$M{-}O + H{-}A \rightarrow MA + H_2O$$

$$CuO(s) + 2HCl(aq) \rightarrow CuCl_2(aq) + H_2O(l)$$

$$CuO(s) + 2CH_3COOH(aq) \rightarrow Cu(CH_3COO)_2(aq) + H_2O(l)$$

Reaction with metal hydroxides

Acids undergo a similar reaction with metal hydroxides to produce a salt and water. The reaction involves:

- The metal cation (M^+) in the hydroxide bonding with the anion (A^-) in the acid to form the salt (MA).
- The hydrogen ion (H^+) in the acid bonding with the hydroxide (OH^-) in the base to form water (H_2O).

The general word equation is given below, together with some specific examples:

$$\text{metal hydroxide} + \text{acid} \rightarrow \text{salt and water}$$
$$M{-}OH + H{-}A \rightarrow MA + H_2O$$

$$NaOH(aq) + HNO_3(aq) \rightarrow NaNO_3(aq) + H_2O(l)$$

$$NaOH(s) + CH_3COOH(aq) \rightarrow NaCH_3COO(aq) + H_2O(l)$$

The salt produced in the reaction depends on the parent acid and base that react (**Figure 2**). For example, the reaction of sodium hydroxide and nitric acid produces sodium nitrate.

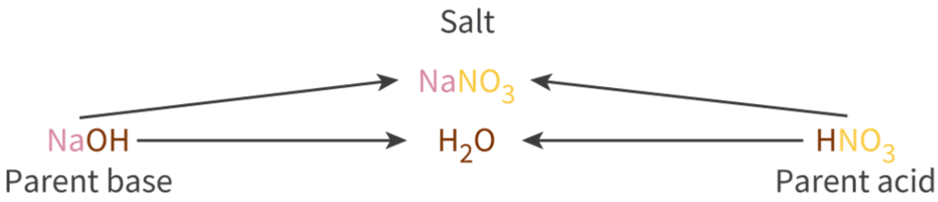

Figure 2. The make-up of a salt is defined by the acid and base it is made from.

More information for figure 2

The diagram illustrates how a salt is formed from the reaction between a parent acid and a parent base. On the left side, the parent base is represented as NaOH, and on the right side, the parent acid is HNO_3. Arrows point from each reactant toward a central product labeled as $NaNO_3$, which is the salt. Below the salt, another arrow points downwards to H_2O, indicating the formation of water. This schematic represents the chemical equation where sodium hydroxide (NaOH) reacts with nitric acid (HNO_3) to produce sodium nitrate ($NaNO_3$) and water (H_2O).

[Generated by AI]

Worked example 1

Determine the products of these reactions and complete the balanced equations.

1. $Mg(OH)_2(aq) + H_2SO_4(aq)$
2. $CaO(s) + HNO_3(aq)$

First identify the substances involved, then write the formulas of the products, then balance the equations.

1. $Mg(OH)_2(aq) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + 2H_2O(l)$

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2. $CaO(s) + 2HNO_3(aq) \rightarrow Ca(NO_3)_2(aq) + H_2O(l)$

Worked example 2

Identify the acid and base that produce the following salts:

1. Li_2SO_4
2. $Mg(NO_3)_2$
3. CH_3CH_2COOK
4. $CaCl_2$

The metal cation is from a base (combine with OH^-) and the anion is from the acid (combine with H^+).

1. LiOH and H_2SO_4
2. $Mg(OH)_2$ and HNO_3
3. KOH and CH_3CH_2COOH
4. $Ca(OH)_2$ and HCl

Reaction with metal carbonates

Acids react with metal carbonates to produce a salt, carbon dioxide gas and water. The reaction involves:

- The metal cation (M^+) in the carbonate bonding with the anion (A^-) in the acid to form the salt (MA).
- The hydrogen (H^+) in the acid together with the carbonate ion forming two products – carbon dioxide (CO_2) and water (H_2O).

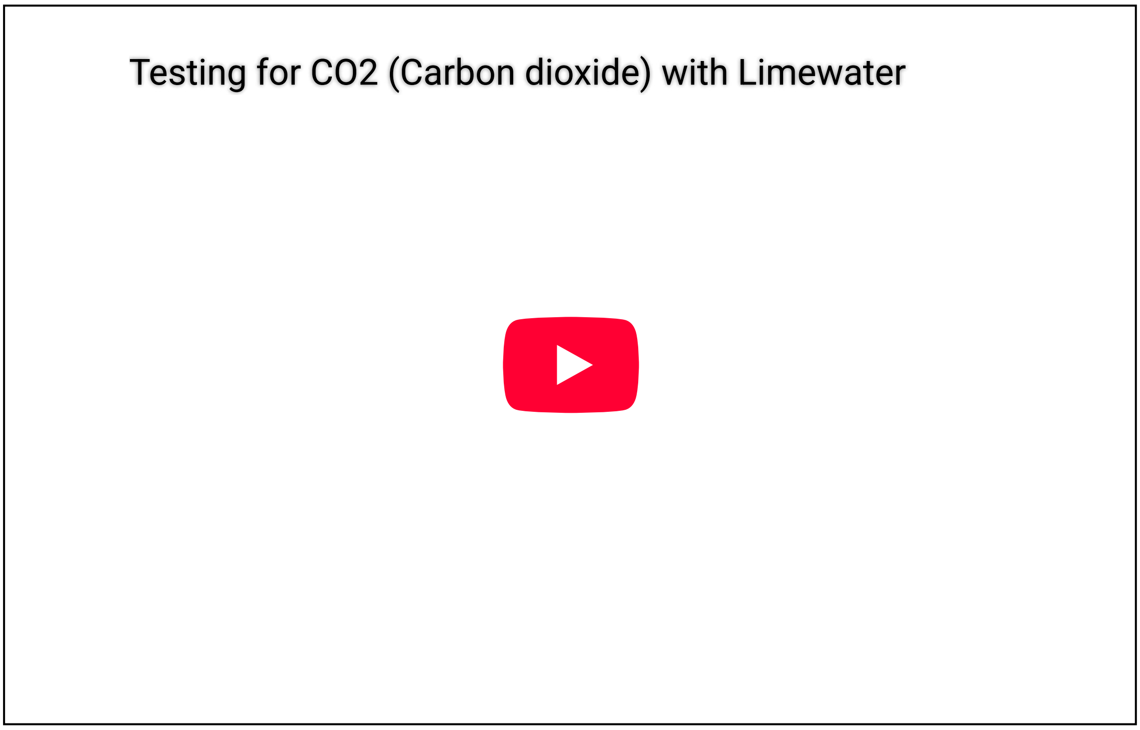
The general word equations are given below, together with some specific examples.

$$\text{metal carbonate} + \text{acid} \dashrightarrow \text{salt} + \text{water} + \text{carbon dioxide}$$
$$M{-}CO_3 + H{-}A \dashrightarrow MA + H_2O + CO_2$$

$$MgCO_3(s) + 2HCl(aq) \dashrightarrow MgCl_2(aq) + H_2O(l) + CO_2(g)$$

$$MgCO_3(s) + 2CH_3COOH(aq) \dashrightarrow Mg(CH_3COO)_2(aq) + H_2O(l) + CO_2(g)$$

The presence of carbon dioxide gas can be tested for by bubbling the gas through an aqueous solution of calcium hydroxide, $Ca(OH)_2(aq)$, otherwise known as limewater. If carbon dioxide gas is present, the limewater turns a ‘milky’ colour as a solid precipitate of calcium carbonate is formed (**Video 1**).



Video 1. Demonstration of the limewater test.

Can you correlate the reaction occurring in the limewater to an acid-base reaction?

Remember that carbon dioxide increases the acidity of water, and limewater is a basic solution.

Worked example 3

Complete and balance the equation for the reaction between solid sodium carbonate and hydrochloric acid:

$$Na_2CO_3(s) + HCl(aq) \dashrightarrow \_\_\_\_\_\_$$

Identify first the type of substances involved, then write their formulas and those of the products, then balance the equations.

$$Na_2CO_3(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$$

International Mindedness

Coral reefs appear to be made of rock, but are actually colonies of animals called polyps. The polyps have a reciprocal relationship with the plant life in the ocean, providing carbon dioxide and waste products which are used for photosynthesis. In return, the plants provide oxygen and nutrients to the polyps, allowing them to make their calcium carbonate skeletons. Carbon dioxide in the air dissolves into the ocean, forming carbonic acid. The steady increase in carbon dioxide that comes from the burning of fossil fuels has caused an increase in the concentration of carbonic acid, which then reacts with the calcium carbonate skeletons in a neutralisation reaction. Globally, this has resulted in a 40—50% decrease in coral over the last 30 years. Coral reefs provide protection to the shoreline, a habitat for marine organisms and limestone for construction. All countries are connected by the air we breathe and the water we share, threatening the survival of coral reefs and the benefits we gain from them.

Reaction with metal hydrogencarbonates

Acids react with metal hydrogencarbonates in a very similar reaction. The reaction involves:

- The metal cation (M^+) in the carbonate bonding with the anion (A^-) in the acid to form the salt (MA).
- The hydrogen ion (H^+) in the acid together with the hydrogencarbonate ion forming two products – carbon dioxide (CO_2) and water (H_2O).

The general word equations are given below, together with some specific examples.

metal hydrogencarbonate + acid ⇢ salt + water + carbon dioxide

$$M{-}HCO_3 + H{-}A \dashrightarrow MA + H_2O + CO_2$$

$$NaHCO_3(s) + HCl(aq) \dashrightarrow NaCl(aq) + H_2O(l) + CO_2(g)$$

$$NaHCO_3(s) + CH_3COOH(aq) \dashrightarrow NaCH_3COO(aq) + H_2O(l) + CO_2(g)$$

Such reactions occur in cave formation in many parts of the world, including in the Waitomo cave in New Zealand. Acidic rain with carbonic acid content reacts with limestone rocks to create beautiful caves (**Figure 3**).

Figure 3. Waitomo cave in New Zealand.

Credit: daboost, Getty Images

Worked example 4

Complete and balance the equation for the reaction between lithium hydrogencarbonate and hydrochloric acid:

$$LiHCO_3(s) + HCl(aq) \dashrightarrow \_\_\_\_\_\_$$

First identify the type of substances involved, then write their formulas and those of the products before balance the equations.

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$$LiHCO_3(s) + HCl(aq) \rightarrow LiCl(aq) + H_2O(l) + CO_2(g)$$

Theory of Knowledge

If you have ever purchased art supplies, you might have noticed the term acid-free on the packaging for paper. Paper is traditionally made from wood pulp containing *lignin*, a fibre that provides strength to the paper, however in time it will break down producing hydrochloric acid. When paper comes in contact with acid, it causes yellow-brown discolouration, known as foxing in the art community. Acid-free paper became commercially available in the 1950s made with lignin removed and infused with calcium hydrogencarbonate or magnesium hydrogencarbonate to neutralise any acids that might come into contact with the paper to prevent foxing.

Paintings and sketches made on traditional paper that have undergone foxing can be restored by neutralising the acids using a base — such as calcium hydroxide or ammonia. Art restoration, however, is a controversial process that risks damaging or altering the piece such that it no longer resembles the original form, making the notion of preservation questionable.

Does knowledge of the chemical process required to save a piece of artwork from damage compel us to use it at the risk of altering the original work?

Reaction with ammonia and amines

Acids undergo neutralisation reactions with aqueous ammonia and amines, which are bases, to produce a salt. The reaction involves:

- The nitrogen contains a lone pair that is capable of accepting a hydrogen so the ammonia accepts a hydrogen ion, forming the ammonium ion, NH_4^+, and the anion from the acid remains unchanged.
- The two oppositely charged species are now attracted to each other electrostatically.

The general word equation is given below, together with some specific examples:

$$\text{ammonia} + \text{acid} \rightarrow \text{salt}$$

$$NH_3 + H{-}A \rightarrow [NH_4]^+[A]^-$$

$$NH_3(aq) + CH_3COOH(aq) \rightarrow NH_4CH_3COO(aq)$$

$$NH_3(aq) + HNO_3(aq) \rightarrow NH_4NO_3(aq)$$

$$\text{amine + acid} \rightarrow \text{salt}$$

$$RNH_2 + H{-}A \rightarrow [RNH_3]^+[A]^-$$

$$CH_3NH_2(aq) + HNO_3(aq) \rightarrow CH_3NH_3NO_3(aq)$$

Worked example 5

Complete and balance the equations for the reaction between:

1. ammonia and hydrochloric acid

$$NH_3(aq) + HCl(aq) \rightarrow \_\_\_\_\_\_$$

2. methyl amine and hydrochloric acid

$$CH_3NH_2(aq) + HCl(aq) \rightarrow \_\_\_\_\_\_$$

Identify first the type of substances involved, then write their formulas and those of the products, then balance the equations.

1. $NH_3(aq) + HCl(aq) \rightarrow NH_4Cl(aq)$
2. $CH_3NH_2(aq) + HCl(aq) \rightarrow CH_3NH_3Cl(aq)$

Practical skills

Tool 1: Experimental techniques — Applying techniques in separating mixtures

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Can you apply the techniques learned in subtopic S1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-40864/) to separate the salts formed in neutralisation reactions?

Making connections

Recall from subtopic R1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43468/) that reactions can be exothermic or endothermic, and that all energy in reactions is stored in bonds. Neutralisation reactions are exothermic. How can this be explained in terms of bond enthalpies?

Try the simulation in the following activity to check your understanding of the properties of acids and bases.

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:**
  - Thinking skills — Being curious about the natural world; Applying key ideas and facts in new contexts
  - Social skills — Working collaboratively to achieve a common goal
- **Time required to complete activity:** 20 minutes
- **Activity type:** Groups of 2—3

Instructions

1. Open the online lab room (https://web.archive.org/web/20221006082549/http://amrita.olabs.edu.in/ola sub=CHE&cat=PHC&exp=ToStudyThePropertiesOfAcidsNBases&tempId=ola IN) on properties of acids and bases.
2. Choose the test with sodium carbonate (Solid Na_2CO_3) and Sample 1.
3. Press Help to get instructions on how to run the test.
4. Record your observations.
5. Conclude if the Sample 1 is acidic or basic. Are you correct in your conclusion?
6. Repeat steps 2 to 5 for samples 1 to 4.

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7. What would the observations be if samples 1 to 4 were mixed with:

    (a) sodium hydroxide
    (b) ammonia
    (c) sodium hydrogencarbonate?